Java Collections Framework Reference

Main Class Name requirement: JavaFirtProgram

This document details the hierarchy of the Java Collection Framework, providing definitions and code examples for each primary interface and class shown in your diagram.

1. List Interface (Ordered Collection)

A collection that maintains insertion order and allows duplicate elements.

- **ArrayList:** Dynamic array implementation.
- **LinkedList:** Doubly-linked list implementation.
- **Vector:** Thread-safe legacy dynamic array.
- **Stack:** LIFO (Last-In-First-Out) implementation.

```java
import java.util.*;

public class JavaFirtProgram {
    public static void main(String[] args) {
        List<String> arrayList = new ArrayList<>();
        arrayList.add("Java");
        arrayList.add("Python");

        List<String> linkedList = new LinkedList<>();
        linkedList.add("Node 1");

        Stack<Integer> stack = new Stack<>();
        stack.push(100);

        System.out.println("ArrayList: " + arrayList);
        System.out.println("Stack Top: " + stack.peek());
    }
}
```

2. Queue & Deque Interface (Processing Order)

Designed for holding elements prior to processing.

- **PriorityQueue:** Orders elements based on natural priority.
- **Deque:** Double-ended queue (insertion/removal at both ends).
- **ArrayDeque:** Faster resizable-array implementation of Deque.

```
import java.util.*;

public class JavaFirtProgram {
    public static void main(String[] args) {
        Queue<Integer> pQueue = new PriorityQueue<>();
        pQueue.add(30);
        pQueue.add(10); // Automatically sorted to head

        Deque<String> deque = new ArrayDeque<>();
        deque.addFirst("Front");
        deque.addLast("Back");

        System.out.println("PriorityQueue Head: " + pQueue.peek());
        System.out.println("Deque State: " + deque);
    }
}
```

3. Set & SortedSet Interface (Unique Collection)

A collection that contains no duplicate elements.

- **HashSet:** No order guarantee; uses hashing.
- **LinkedHashSet:** Maintains insertion order.
- **TreeSet (SortedSet):** Elements stored in a sorted tree structure.

```java
import java.util.*;

public class JavaFirtProgram {
    public static void main(String[] args) {
        Set<String> hashSet = new HashSet<>();
        hashSet.add("Apple");
        hashSet.add("Apple"); // Duplicate: will not be added

        SortedSet<Integer> treeSet = new TreeSet<>();
        treeSet.add(50);
        treeSet.add(10);
        treeSet.add(20);

        System.out.println("HashSet (Unique): " + hashSet);
        System.out.println("Sorted TreeSet: " + treeSet);
    }
}
```